

A Machine Learning and Deep Learning Integrated Model to Detect Criminal Activities

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Abstract—Crime is considered an intended execution of an act that is harmful to society and punishable under criminal law. Bangladesh as a developing country has experienced severe crime-related issues in recent years due to the constantly growing criminal activities. The law enforcement sector of Bangladesh is dedicated to implementing the law and reducing fear of offense to make Bangladesh a secure place to live. Despite this, the traditional ways of detecting crimes in Bangladesh are currently slow and less effective because they are not getting enough technological support. As a result, most of the time the crimes get reported after the occurrence. So to attain efficiency in policing activities it has become significant to develop such a system that can notify the police department before the occurrence of a crime. In our proposed model, we develop a system that can prevent crime by detecting and notifying real-time criminal activities beforehand. To carry out this concept, we build a central system consisting of three modules i.e. criminal activity identification module, weapon detection module, and criminal detection module. Each of the modules is centrally connected and can give an alert when any type of abnormality occurs. We use Deep Learning and Machine Learning algorithms to perform criminal activity detection and face recognition of criminals. For each of the modules, we collect and prepare our own data set. We perform various experiments to investigate the effectiveness of the method. The evaluation result shows that our proposed method performs impressively in identifying criminal activities and criminals.

Index Terms—Criminal Activities, Law Enforcement, Personal-ity Analysis, Face Recognition, Deep Learning, Machine Learning

I. INTRODUCTION

Bangladesh evolved independently on 16 December 1971 and is now considered a developing country. Crime has been a crucial problem from the beginning of the country's independence [1]. Different economic factors like violence, unemployment, and injustice cause crime here. At present, the crime rate of Bangladesh is around 69.94%, which is very threatening¹. However, most of the time, the crimes remain unreported or get reported after the occurrence, which induces risks and challenges for the police to attain efficiency in their policing activities. To overcome the risks and facilitate the work of the law enforcement sectors of Bangladesh, it has become significant to develop a system that can notify them

whenever a crime occurs. It would be quite helpful if a smart system could detect real-time crime by itself and alert the police immediately about the crime scene and the probable suspects.

To our best knowledge, no such system exists in Bangladesh to detect or forecast crime in real-time and warn high authorities to prevent further damage. However, recent advances in machine Learning (ML) and deep learning (DL) algorithms would be excellent support for this because these algorithms provide effective results in real-time by analyzing and processing several parameters. So, in our proposal, we develop a core system that can identify real-time criminals and detect criminal activities to prevent crime in Bangladesh. We use both approaches of ML and DL to analyze data sets and detect crime in real-time. It will also assist law enforcement authorities in receiving early warning of illegal activities and allow them to take appropriate action on time.

In this paper, we propose three significant modules of a pipeline i.e. criminal activity detection, weapon detection, and criminal prediction. Each of the modules works independently and sends notifications to the central system. For the criminal activity detection module, we analyze the UCF-Crimes dataset to detect crime through video surveillance cameras. We use the Long-term Recurrent Convolutional Network (LRCN) [2], a deep learning model to detect criminal activities like explosion, stealing, robbery, assault, fighting, road accidents, and burglary. For the weapon detection module, we use the Pashtaw dataset to detect weapons through video surveillance cameras. We use a real-time object detector model YOLOv7 [3] to detect weapons. We create a criminal database by ourselves for the criminal forecasting module with their wanted status. For criminal detection, we use dlib's state-of-the-art deep learning model Face Recognition² to detect the face of a criminal from the database along with basic information. Then we examine the social media status of the suspected person to forecast his personality. To analyze social media accounts, we use TextBlob [4]. In the result analysis section, we perform different experiments to investigate the effectiveness of the System.

¹https://www.numbeo.com/crime/country/_result.jsp?country=Bangladesh

²https://github.com/ageitgey/face_recognition

The rest of the paper is organized as follows. In Section II, related work on this topic has been discussed along with some background data. Section III represents details of the proposed model. The analysis and outcomes of the experiment are shown in Section IV. The conclusionV contains shortcomings and future work of the proposed model.

II. RELATED WORK

In recent years, machine learning and deep learning algorithms have been widely used in the crime detection field. In [5], the authors demonstrated the effectiveness of ML algorithms in predicting crime patterns and other applications, such as identifying criminal hotspots, developing criminal profiles, and identifying criminal trends using WEKA. Kim et.al [6] used ML predictive models K-Nearest Neighbour (KNN) and boosted decision tree to predict crime. A total of 560,000 crime datasets between 2003 and 2018 were analyzed with an accuracy of 39% to 44%. In [7], the prediction of crime-related statistics in Philadelphia and Pennsylvania was discussed using ML techniques like logistic regression, KNN, ordinal regression, and tree techniques with 69% accuracy.

In [8], the authors examined a dataset made up of several crimes and made predictions about the kinds of crimes that would happen soon using models including KNN classification, logistic regression, decision trees, random forests, support vector machine (SVM), and Bayesian approaches with an accuracy of roughly 78%. Prithi et.al [9] showed how to anticipate crime rates utilizing a graphical user interface and ML algorithms. The authors of [10] and [11] employed KDE and social media data to predict crime. In [12] they used a deep neural network (DNN)-based feature-level data fusion method to forecast the occurrence of crime with accuracies of 67.01% and 66.33%, respectively. In [13] random forest, decision tree, and k-nearest neighbor (KNN) algorithms were used to forecast criminal conduct with 99.16% accuracy.

The decision tree (J48) algorithm was taken into consideration in [14] when creating a crime prediction model. In [15] a brand-new method was applied to detect crime scene video surveillance systems in real-time violence detection using the spatiotemporal (ST) technique with a classification method based on Deep Reinforcement Neural Networks (DRNN). In [16] random forest, naïve Bayes, and SVM were used to forecast the frequency of crimes and hotspots with an accuracy of 75.6% and 65.3%. In [17] future crime patterns in Bangladesh are predicted using a linear regression model. The work cited in [18] listed a number of the analysis and forecasting of various crime reports using data from the Bangladesh Police Dataset and the ACLED Dataset. Obuandike et al. [19] attempted to employ the ZeroR algorithm and a decision tree in their study but were unable to get an accuracy of more than 60%.

III. PROPOSED WORK

The main target of our research is to develop a core system that can alert crime fighters immediately after detecting real-time criminal activities or criminals through video surveillance

cameras. For this purpose, we propose a pipeline integrated with three modules i.e. criminal activity detection, weapon detection, and criminal prediction module. Each of the modules connects centrally and give alert to the high authorities to prevent crime. The following subsections contain detailed descriptions of the proposed three modules.

A. Criminal Activity Detection

To detect criminal activities, we collect UCF-Crimes³, a large-scale dataset that consists of 1900 real-world surveillance videos of 13 abnormalities. After analyzing the videos we collected 1000 videos of 7 different abnormalities, which are more common in Bangladesh, and built a new dataset. To obtain a more accurate and clean cut of the crime scene, we perform different data pre-processing such as video resizing using KAPWING⁴, a modern video editing tool. We split the data into train and test groups using a 75:25 ratio. After collecting video data, we build an LRCN model using Tensorflow [20]. The LRCN model combines two prominent deep learning algorithms CNN [21] and LSTM [21] to provide more accuracy. The process flow of the criminal activity detection module is depicted in Figure 1.

B. Weapon Detection

For our weapon detection module, we use the dataset Pashtaw⁵, which contains 3,000 images of labeled pistols. We pre-process the dataset for the best fit of our model. We have cloned our dataset with YOLOv7, a real-time object detector with computer vision techniques to detect weapons in real time. We split the data into train and test groups using an 80:20 ratio. Figure 2 represents the workflow of the weapon detection technique.

C. Criminal Prediction

In this module, we detect the face of a criminal through video surveillance cameras along with the basic information from a criminal database. If the criminal is wanted, our system gives alerts immediately. If no information is available, then we analyze the social media status of the detected face to analyze the personality. We divide this module into two subsections i.e. criminal detection, and sentiment analysis.

1) *Criminal Detection*: As we can not collect any central database of criminals, we create our dataset, which contains images and their basic information like name, address, wanted status, ETC. For face recognition in real-time, we use the face recognition library from dlib's state-of-the-art model with deep learning algorithms. From the model, we use the Histogram of Oriented Gradients (HOG)+Linear SVM face detector technique⁶. After detecting any criminal, we check it on the criminal database to collect information about the criminals. Figure 3 represents the workflow of the face detection technique.

³<https://www.kaggle.com/datasets/odins0n/ucf-crime-dataset>

⁴<https://www.kapwing.com/video-editor>

⁵<https://github.com/topics/weapons-detection>

⁶shorturl.at/ePRS9

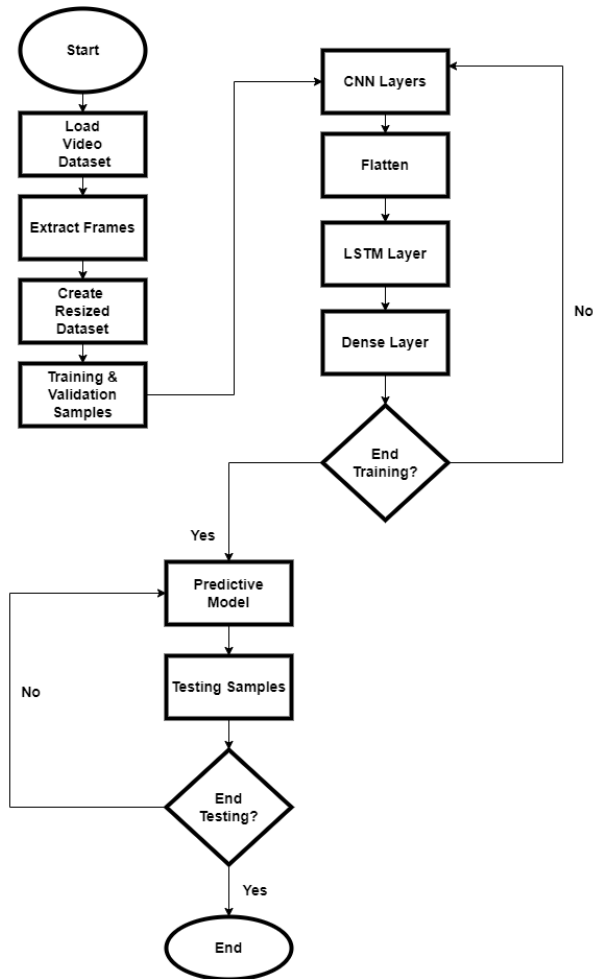


Fig. 1. Criminal Activity Detection Method

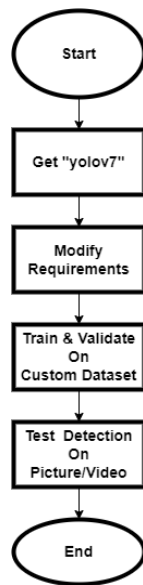


Fig. 2. A Systematic Approach for Weapon Detection

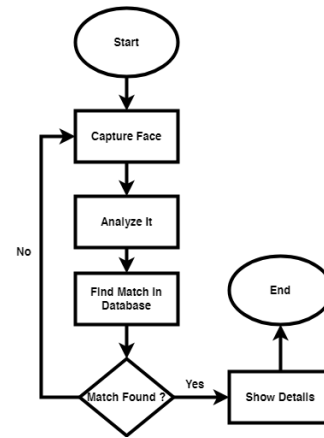


Fig. 3. A Systematic Approach for Face Detection

2) *Sentiment Analysis*: To add more clarity to our work, we analyze and predict sentiment based on the social media status of detected persons. We also create a dataset by ourselves. We use Tweepy [22] to access the Twitter API and collect tweets of selected users. As we collect text data, this may contain different noises. We perform some preprocessing, such as removing usernames, patterns, @, hashtags, and short words, splitting the white space, replacing digits with spaces, and removing punctuation, numbers, and special characters.

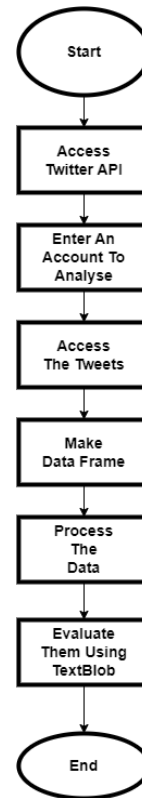


Fig. 4. A Systematic Approach for Sentiment Analysis

We use TextBlob to dive into the natural language process-

ing (NLP)⁷ task for sentimental analysis. We evaluate the data using a Scatter Plot of Subjectivity-Polarity, a Bar Chart of Sentiment, and a World Cloud view for the most used words in all Tweets to anticipate any crime that may be committed in the near future. Figure 4 represents the workflow of the sentiment analysis technique.

IV. EXPERIMENTAL RESULT AND ANALYSIS

The following subsections contain a detailed description of the results obtained from different experiments.

A. Criminal Activity Detection

As mentioned, we collect a dataset consisting of 1,000 videos of 7 abnormalities from the UCF-Crimes dataset. Among them, 174 videos are assault-related, 120 are explosion-related, 196 are stealing-related, 198 are robberies, 184 are road accidents, and 144 are burglaries videos. We train the model using algorithms like CNN and LSTM. Figure 5 represents video samples of criminal activities captured by security cameras from the collected dataset.



Fig. 5. Criminal Activity Detection Test Cases



Fig. 6. Total Loss VS Total Validation Loss

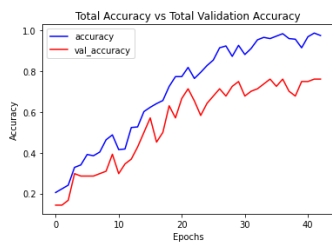


Fig. 7. Total Accuracy VS Total Validation Accuracy

Here, Figure 6 represents the training and validation loss of the model. Figure 7 represents the model accuracy curve where training accuracy is slightly greater than validation accuracy. We achieved an accuracy of 84% for the criminal activity detection task, which is quite impressive.

⁷<http://dl.acm.org/citation.cfm?id=2078183.2078186>

B. Weapon Detection

As mentioned earlier, for weapon detection, we use the Pashtaw dataset. Among the 3000 images of labeled pistols, we collected 1120 images for this task. Figure 8 represents a sample of a weapon detection dataset where a short gun has been detected using YOLOv7.



Fig. 8. Weapon Detection Test Case

To evaluate this module, we calculate the F1-Curve, PR-Curve, and Confusion Matrix, which are shown in Figure 9, 10, and 11, respectively. The confidence rating that maximizes precision and recall based on Figure 9 is 0.360. The F1 value for this model is about 0.8 and is similar to the distance from the maximum value of 0.8. We choose a confidence level of 0.78 for Figure 10 as a suitable point.

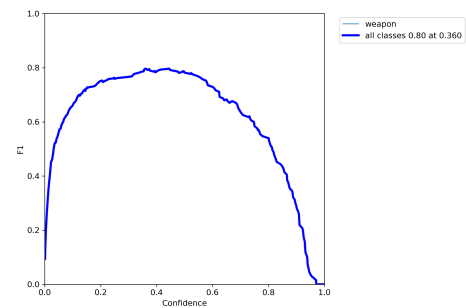


Fig. 9. F1 Score Curve for a Custom YOLOv7 Weapon Detection

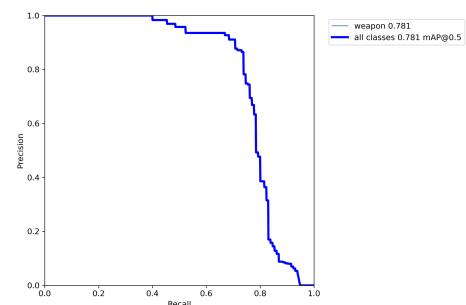


Fig. 10. PR-Curve for Weapon Detection

From these figures, we attain precision value 1, which is the best value, recall 0.8, and F1-Score 0.88, which are pretty

for the most used words in All Tweets is also shown in Figure 16.

Here, in Figure 14, we can see that the majority of the dots are on the positive or neutral(Polarity=0) side of the line, indicating that the user's social media posts are safe. On the other hand, the negativity line is around 20 in Figure 15, indicating that there is a low likelihood of the user committing a crime shortly. Lastly, in Figure 16, no harmful or negative words are seen.

V. CONCLUSION AND FUTURE WORK

Artificial intelligence (AI) is gradually but steadily becoming a capable tool for punishing criminals and preventing criminal conduct. It is no longer just a speculative notion. AI is now being used by law enforcement agencies to improve the productivity of their officers. Our proposed model can be used for surveillance, such as monitoring a crowd for anomalies, analyzing video footage for criminal activity, and applying facial recognition or weapon recognition to better advantage. This policing pipeline is expected to result in changes in security and assurance for society. The combination of our pipeline with current AI programs may prove to be quite beneficial given the rising crime rate. To a greater extent, this technology can aid in the prediction and detection of crime as well as criminals.

In our proposed model, we propose a core system consisting of three modules i.e. criminal activity detection, weapon detection, and criminal forecasting to prevent crime. To achieve high accuracy, we mix deep learning and machine learning techniques. For the sake of our research, we have gathered and examined real-world datasets that include surveillance videos. We employ LSTM and CNN algorithms based on ARNN to detect criminal activities, and for weapon detection, we use YOLOv7. We use Dlib's HOG and Linear SVM to identify criminals by recognizing their faces. In addition to identifying the criminal, we look at their social media status to look forward to potential criminal activities. To analyze the effectiveness of our suggested model, we run considerable examinations of our datasets and evaluate the performance of each module. The experimental study and findings demonstrate that our strategy produces good outcomes in each part.

Even though we have accomplished our objectives, there is still much room for improvement in our core system modules. In the future, by accessing different social media platforms, we can extend our social media analysis using images, videos, check-ins, comments, etc. Moreover, we can use geolocation to predict criminal hotspots and study criminals' body language before committing a crime to prevent crime.

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